Lecture 2

# Constraints and Lagrange's equations

**Objectives.** *Lecture 1 gave the state and the flow of a system whose particles move freely. Rods, hinges and rigid bodies constrain them: the allowed configurations form a* surface *in the space of particle positions. This lecture asks what Newton's law becomes on such a surface. It says when constraint equations define one, what its velocities are and which forces a constraint may exert, and proves that Newton's law with those forces is equivalent to* Lagrange's equations *for $\mathcal{L} = T - V$, in any coordinates on the surface, with $(q, \dot{q})$ as the state. The pendulum, now with its rod, is the example.*

## 2.1 Holonomic constraints

### Constraints

From now on $n$ is reserved for the number of *degrees of freedom*, which constraints make smaller than in Lecture 1. We write $r \in \mathbb{R}^N$ for the list of all Cartesian coordinates of all particles, and $m_i > 0$ for the mass carried by $r_i$ (each particle's mass once per Cartesian component). A rod of length $\ell$ joining particles 1 and 2 forces $|\mathbf{r}_1 - \mathbf{r}_2|^2 = \ell^2$. A *holonomic constraint* is an equation on the positions alone:

$$f_1(r) = \cdots = f_k(r) = 0, \qquad f = (f_1, \dots, f_k) \colon U \to \mathbb{R}^k, \tag{2.1}$$

with $U \subseteq \mathbb{R}^N$ open and $f$ of class $C^2$. The *configuration space* is the set $Q = \{r \in U : f(r) = 0\}$.

Not every equation cuts out a set fit for calculus: $\{xy = 0\}$ is two crossing lines, $\{x^2 + y^2 = z^2\}$ a cone, and at the bad point the gradient vanishes.

**Definition 2.1** (Regular value)**.** Let $f \colon U \to \mathbb{R}^k$ be $C^2$ on an open $U \subseteq \mathbb{R}^N$. A value $c \in \mathbb{R}^k$ is *regular* if the $k \times N$ matrix $Df(r)$ has rank $k$ at every $r$ with $f(r) = c$; equivalently, $\nabla f_1(r), \dots, \nabla f_k(r)$ are linearly independent there. The constraints (2.1) are *regular* if 0 is a regular value.

**Proposition 2.2** (Regular level sets are locally graphs)**.** *Let $c$ be a regular value of $f \colon U \to \mathbb{R}^k$, $Q = \{f = c\}$ and $n = N - k$. For every $r_0 \in Q$ the coordinates of $\mathbb{R}^N$ can be numbered so that $r = (y, z)$, $y \in \mathbb{R}^n$, $z \in \mathbb{R}^k$, and there are open sets $V \subseteq \mathbb{R}^n$, $Z \subseteq \mathbb{R}^k$ with $r_0 \in V \times Z$ and a $C^2$ map $g \colon V \to Z$ with*

$$Q \cap (V \times Z) = \{(y, g(y)) : y \in V\}.$$

*If $f$ is $C^p$, so is $g$.*

*Proof.* $Df(r_0)$ has rank $k$, so some $k$ of its columns are independent; call the corresponding coordinates $z$. Then $\partial f / \partial z(r_0)$ is invertible, and the implicit function theorem ($C^p$ data, $C^p$ solution) gives $V$, $Z$ and $g$. ∎

**Definition 2.3** (Surface, local parametrisation, generalised coordinates)**.** A set $Q \subseteq \mathbb{R}^N$ is a *surface of dimension* $n$ if every point of $Q$ lies in the image of a *local parametrisation*: a $C^2$ map $r\colon V \to \mathbb{R}^N$ on an open $V \subseteq \mathbb{R}^n$ such that

1. $r$ is a bijection of $V$ onto $Q \cap W$ for some open $W \subseteq \mathbb{R}^N$;
2. the $N \times n$ matrix $Dr(q)$ has rank $n$ for every $q \in V$;
3. the inverse map $q\colon Q \cap W \to V$ is continuous.

The components of $q$ are *generalised coordinates* on the *patch* $Q \cap W$; $n = \dim Q$ is the number of degrees of freedom.

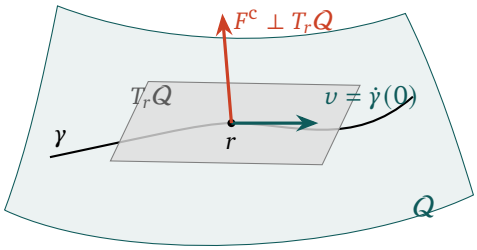



Figure 2.1: A constraint surface $Q$, a curve $\gamma$ in it, and its tangent plane at $r$: the allowed velocities $T_rQ = \ker Df(r)$ (§ 2.3). An ideal constraint force (§ 2.4) is normal to it.

**Corollary 2.4.** *If $c$ is a regular value of $f\colon U \to \mathbb{R}^k$, then $\{f = c\}$ is a surface of dimension $N - k$ (Figure 2.1).*

*Proof.* $r(y) = (y, g(y))$ from Proposition 2.2, with $W = V \times Z$, has $Dr = \left(\begin{smallmatrix}\mathbb{1}\\ Dg\end{smallmatrix}\right)$ of rank $n$ and the continuous inverse $(y, z) \mapsto y$. ∎

Regularity concerns the equations, not the set: $(x^2 + y^2 - 1)^2 = 0$ is the circle with zero gradient. Condition (3) is needed too. The map $r(s) = (\sin 2s, \sin s)$ on $(-\pi, \pi)$ is a bijection onto a figure-eight, with $r'(s) \neq 0$, but its inverse jumps at the crossing $r(0) = (0, 0)$: points of the curve near the origin come from $s$ near 0 and from $s$ near $\pm\pi$. Condition (3) is what the next lemma uses.

**Lemma 2.5** (Changes of coordinates · *sketched*)**.** *Let $r\colon V \to \mathbb{R}^N$ and $\tilde r\colon \tilde V \to \mathbb{R}^N$ be local parametrisations of a surface $Q$, with inverses $q$ and $\tilde q$. On the open set $V_0 = r^{-1}(\tilde r(\tilde V))$ the* transition map *$\varphi = \tilde q \circ r$, for which $r = \tilde r \circ \varphi$, is $C^2$ with invertible derivative.*

*Proof.* Near $\tilde q_0 = \varphi(q_0)$ some $n$ Cartesian coordinates $\pi\colon \mathbb{R}^N \to \mathbb{R}^n$ make $D(\pi \circ \tilde r)(\tilde q_0)$ invertible, so $\pi \circ \tilde r$ has a $C^2$ local inverse $h$ (inverse function theorem). By condition (3), $\varphi(q)$ stays in the domain of $h$ for $q$ near $q_0$, and there $\varphi = h \circ \pi \circ r$ is $C^2$. So is $\varphi^{-1} = q \circ \tilde r$; hence $D\varphi$ is invertible. ∎

## 2.2 Configuration spaces

### The pendulum: a circle

A bob on a rod of length $\ell$ in a vertical plane: $f = x^2 + y^2 - \ell^2$, $\nabla f \neq 0$ on the circle, so $Q = S^1$, $n = 1$. The angle from the downward vertical, $r(\theta) = \ell(\sin\theta, -\cos\theta)$, is a local

parametrisation on $(-\pi, \pi)$, missing the top; a second one on $(0, 2\pi)$ misses the bottom. The transitions are $\theta \mapsto \theta$ and $\theta \mapsto \theta + 2\pi$; in practice $\theta$ is used modulo $2\pi$ (§ 1.5).

### The rigid body: SO(3)

A rigid body not contained in a line, with a point fixed at the origin, is placed by the matrix $R$ carrying a reference placement to the current one (two independent body directions fix $R$). $R$ preserves distances, so it is orthogonal, and a motion from $\mathbb{1}$ cannot jump to $\det R = -1$. Here $N = 9$ and

$$f(R) = R^{\mathsf{T}} R, \qquad Df(R)[X] = R^{\mathsf{T}} X + X^{\mathsf{T}} R,$$

with values in the symmetric matrices: six equations, not nine. The value $\mathbb{1}$ is regular, since $X = \frac{1}{2} RS$ has $Df(R)[X] = S$ for symmetric $S$. So the orthogonal matrices form a surface of dimension $9 - 6 = 3$, and SO(3) is the part of it where the continuous function $\det = \pm 1$ is $+1$.

## 2.3 Velocities

**Definition 2.6** (Tangent space)**.** Let $Q$ be a surface and $r \in Q$. The *tangent space* $T_r Q$ is the set of velocities $\dot\gamma(0)$ of $C^1$ curves $\gamma : (-\varepsilon, \varepsilon) \to \mathbb{R}^N$ with values in $Q$ and $\gamma(0) = r$.

**Theorem 2.7** (Tangent space of a regular level set)**.** *Let $c$ be a regular value of $f : U \to \mathbb{R}^k$, $Q = \{f = c\}$, $r_0 \in Q$, and let $r(q)$ be a local parametrisation with $r(q_0) = r_0$. Then*

$$T_{r_0} Q = \ker Df(r_0) = \operatorname{span}\left\{ \frac{\partial r}{\partial q_1}(q_0), \dots, \frac{\partial r}{\partial q_n}(q_0) \right\}, \tag{2.2}$$

*a linear subspace of $\mathbb{R}^N$ of dimension $n = N - k$, and the vectors $\partial r / \partial q_j (q_0)$ are a basis of it.*

*Proof.* If $\gamma$ is a curve in $Q$ through $r_0$, then $f(\gamma(s)) = c$ for all $s$, and differentiating at $s = 0$ gives $Df(r_0)\dot\gamma(0) = 0$: so $T_{r_0} Q \subseteq \ker Df(r_0)$. Conversely, for $w \in \mathbb{R}^n$ the curve $s \mapsto r(q_0 + sw)$ lies in $Q$ and has velocity $Dr(q_0)w$: so the image of $Dr(q_0)$ lies in $T_{r_0} Q$. Hence

$$\operatorname{im} Dr(q_0) \subseteq T_{r_0} Q \subseteq \ker Df(r_0).$$

The left space has dimension $n$ (condition (2)), and the right one has dimension $N - k = n$ by rank–nullity. So all three are equal, and the $n$ columns of $Dr(q_0)$, which span a space of dimension $n$, are a basis. ∎

For the rigid body, $\ker Df(R) = \{X : R^{\mathsf{T}} X \text{ antisymmetric}\}$: the velocities at $R$ are $\dot R = RA$ with $A^{\mathsf{T}} = -A$, three numbers, the components of the angular velocity in the body.

### Generalised velocities and their transformation law

A motion in the patch is $r(t) = r(q(t))$, and by the chain rule

$$\dot r = \sum_{j=1}^{n} \frac{\partial r}{\partial q_j} \dot q_j, \tag{2.3}$$

so the *generalised velocities* $\dot q_j$ are the components of $\dot r \in T_r Q$ in the basis $\partial r / \partial q_j$. In other generalised coordinates $\tilde q = \varphi(q)$ (Lemma 2.5), $\tilde q(t) = \varphi(q(t))$ and

$$\dot{\tilde q} = D\varphi(q)\, \dot q. \tag{2.4}$$

This is the *transformation law of velocities*; Proposition 4.1 derives from it the law of momenta, which is different.

### The velocity phase space

A state is a configuration with an allowed velocity: a point of

$$TQ = \{(r, v) : r \in Q,\ v \in T_rQ\} \subseteq \mathbb{R}^{2N}.$$

It is described by $f(r) = 0$, $Df(r)v = 0$, and the derivative of $(r, v) \mapsto (f(r), Df(r)v)$ is block triangular, $\left(\begin{smallmatrix} Df & 0 \\ * & Df \end{smallmatrix}\right)$, of rank $2k$: $TQ$ is a surface of dimension $2n$ (if $f$ is $C^3$), with coordinates $(q, \dot q) \mapsto (r(q), Dr(q)\dot q)$. For the pendulum, $TS^1$ is the cylinder of § 1.5.

## 2.4 Ideal constraints and d'Alembert

A constraint is enforced by a force, such as the tension in a rod. Newton's law for the constrained system reads

$$m_i \ddot r_i = F_i + F_i^{\mathrm{c}} \qquad (i = 1, \dots, N), \tag{2.5}$$

with $F$ the applied force and $F^{\mathrm{c}}$ the *constraint force*. The unknowns are $n$ functions for the motion in $Q$ and $N$ for $F^{\mathrm{c}}$, against $N$ equations: without a law for $F^{\mathrm{c}}$ the motion is not determined.

**Definition 2.8** (Virtual displacement, ideal constraint)**.** A *virtual displacement* at $r \in Q$ is a vector $\delta r \in T_rQ$: a displacement the constraint allows. The constraint is *ideal* if the constraint force does no virtual work:

$$F^{\mathrm{c}} \cdot \delta r = 0 \qquad \text{for every } \delta r \in T_rQ. \tag{2.6}$$

Together, (2.5) and (2.6) say $\sum_i (m_i \ddot r_i - F_i)\,\delta r_i = 0$ for every virtual displacement: *d'Alembert's principle*.

This is an axiom about the constraint, not a theorem, and it is what "frictionless" means. By Theorem 2.7 it says $F^{\mathrm{c}}$ is orthogonal to $\ker Df$, which is the span of the rows of $Df$ (linear algebra: $(\ker A)^\perp = \operatorname{im} A^{\mathsf T}$). So

$$F^{\mathrm{c}} = \sum_{\alpha=1}^{k} \lambda_\alpha \nabla f_\alpha \tag{2.7}$$

for some numbers $\lambda_\alpha$, the *Lagrange multipliers*: an ideal constraint force is normal to $Q$ (Figure 2.1). For a rod joining particles 1 and 2, $\nabla |\mathbf{r}_1 - \mathbf{r}_2|^2 = \big(2(\mathbf{r}_1 - \mathbf{r}_2),\ 2(\mathbf{r}_2 - \mathbf{r}_1)\big)$: the forces on the two ends are equal, opposite and along the rod. Friction is not of this form.

Along an actual motion $\dot r \in T_rQ$, so an ideal constraint force does no work on the motion either: $F^{\mathrm{c}} \cdot \dot r = 0$.

## 2.5 Lagrange's equations, and the pendulum

Let the applied force be $F = -\nabla V(r, t)$, the constraints $f(r) = 0$ be $C^2$ and regular, and $r(q)$ a local parametrisation of $Q$ (Proposition 2.2 supplies one near each point). Call

$v(q,\dot q) = Dr(q)\,\dot q$ the right-hand side of (2.3), with $\dot q$ an independent variable, and define the *Lagrangian*

$$T(q,\dot q) = \tfrac12 \sum_{i=1}^{N} m_i\, v_i(q,\dot q)^2, \qquad V(q,t) = V\big(r(q),t\big), \qquad \mathcal L = T - V. \tag{2.8}$$

**Lemma 2.9** (Two identities). *With $v$ as above,*

$$(a)\quad \frac{\partial v}{\partial \dot q_j} = \frac{\partial r}{\partial q_j}, \qquad (b)\quad \frac{\mathrm d}{\mathrm dt}\Big[\frac{\partial r}{\partial q_j}\big(q(t)\big)\Big] = \frac{\partial v}{\partial q_j}\big(q(t),\dot q(t)\big)$$

*for every $C^2$ curve $q(t)$.*

*Proof.* (a): $v$ is linear in $\dot q$, with coefficient $\partial r/\partial q_j$ on $\dot q_j$. (b): both sides equal $\sum_l \frac{\partial^2 r}{\partial q_l \partial q_j}\dot q_l$, the left by the chain rule, the right by differentiating $v$ in $q_j$; mixed partial derivatives of a $C^2$ map commute. ∎

**Theorem 2.10** (Lagrange's equations). *Under the hypotheses above, let $q(t)$ be a $C^2$ curve in the parameter domain and $r(t) = r(q(t))$. The following are equivalent:*

1. (Newton with an ideal constraint.) *$m_i \ddot r_i = -\partial V/\partial r_i + F^{\mathrm c}_i$ with $F^{\mathrm c}(t) \perp T_{r(t)}Q$ for all $t$;*
2. (Lagrange.) *for $j = 1,\dots,n$,*

$$\frac{\mathrm d}{\mathrm dt}\frac{\partial \mathcal L}{\partial \dot q_j} - \frac{\partial \mathcal L}{\partial q_j} = 0. \tag{2.9}$$

*In either case the constraint force is determined by the motion: $F^{\mathrm c} = m\ddot r + \nabla V$, componentwise.*

*Proof.* For any $C^2$ curve, set $P_j = \sum_i \big(m_i \ddot r_i + \partial V/\partial r_i\big)\,\partial r_i/\partial q_j$, the component of the residual $m\ddot r + \nabla V$ along the $j$th basis vector. Since $\dot r = v$, and by (a) and then (b),

$$\sum_i m_i \ddot r_i \frac{\partial r_i}{\partial q_j} = \frac{\mathrm d}{\mathrm dt}\Big(\sum_i m_i v_i \frac{\partial v_i}{\partial \dot q_j}\Big) - \sum_i m_i v_i \frac{\partial v_i}{\partial q_j} = \frac{\mathrm d}{\mathrm dt}\frac{\partial T}{\partial \dot q_j} - \frac{\partial T}{\partial q_j}.$$

By the chain rule $\sum_i \frac{\partial V}{\partial r_i}\frac{\partial r_i}{\partial q_j} = \frac{\partial V}{\partial q_j}$, and $V$ does not depend on $\dot q$. Hence

$$P_j = \frac{\mathrm d}{\mathrm dt}\frac{\partial \mathcal L}{\partial \dot q_j} - \frac{\partial \mathcal L}{\partial q_j}.$$

(1)⇒(2): the residual is $F^{\mathrm c}$, which is orthogonal to each $\partial r/\partial q_j \in T_r Q$; so $P_j = 0$. (2)⇒(1): define $F^{\mathrm c} = m\ddot r + \nabla V$. Then $F^{\mathrm c}\cdot \partial r/\partial q_j = P_j = 0$ for all $j$, and by Theorem 2.7 the $\partial r/\partial q_j$ span $T_r Q$; so $F^{\mathrm c} \perp T_r Q$. ∎

So Lagrange's equations are Newton's law dotted with the basis vectors $\partial r/\partial q_j$, and the unknown force drops out because the axiom makes it orthogonal to all of them. The theorem holds in *every* patch; on an overlap both sets of equations describe the same motions, since both are equivalent to (1). Why their *form* survives any change of coordinates, with no Newton in sight, is Lecture 3's subject.

**Corollary 2.11** (Determinacy on the constraint surface). *Write $E = \partial r/\partial q$ ($N \times n$) and $G = E^{\mathsf T}\,\mathrm{diag}(m_i)\,E$. Then $G$ is positive definite, and (2.9) can be solved for $\ddot q$:*

$$\ddot q = G(q)^{-1}\,\Phi(q,\dot q,t)$$

> *for a function $\Phi$ built from $r$, $V$ and their derivatives. If $r$ and $V$ are $C^3$, this is a $C^1$ vector field on the coordinates $(q,\dot q)$ (which fix the velocity through (2.3)), and T1 of Lecture 1 (Theorem 1.3: unique maximal solutions) applies: $(q,\dot q)$ at a time $t_0$ determines the motion for as long as it stays in the patch, and with it the constraint force.*

*Proof.* $\dot q^{\mathsf T} G\dot q = \sum_i m_i (E\dot q)_i^2 > 0$ for $\dot q \neq 0$, as $E$ has rank $n$. $T$ is quadratic in $\dot q$ with Hessian $G$, so $\partial\mathcal{L}/\partial\dot q = G\dot q$ and (2.9) reads $G\ddot q = \Phi$. $\Phi$ contains second derivatives of $r$, $C^1$ when $r$ is $C^3$; the parametrisations of Proposition 2.2 are $C^3$ when $f$ is. If $V$ depends on $t$, the field is made autonomous as in (1.6). ∎

The axiom of ideal constraints restores determinacy: for the $C^3$ data of Corollary 2.11, the state $(q,\dot q)$ determines a unique maximal motion, and the constraint force follows from it.

**Example 2.12** (No constraint: curvilinear coordinates)**.** With $k=0$ the theorem is Newton's law in arbitrary coordinates. A free particle in the plane in polar coordinates, $\rho>0$, has $\mathcal{L} = \frac12 m(\dot\rho^2 + \rho^2\dot\varphi^2)$, and (2.9) gives

$$m\ddot\rho = m\rho\dot\varphi^2, \qquad \frac{\mathrm d}{\mathrm dt}\bigl(m\rho^2\dot\varphi\bigr) = 0,$$

valid until the path reaches the origin. The centrifugal term is the price of curved coordinates, not a force.

### The pendulum

The recipe: choose coordinates, write $r(q)$, express $T$ and $V$ in $(q,\dot q)$, and write (2.9).

With $r = \ell(\sin\theta, -\cos\theta)$ and gravity $-mg$ in the second coordinate, $T = \frac12 m\ell^2\dot\theta^2$ and $V = mgy = -mg\ell\cos\theta$, so

$$\mathcal{L} = \tfrac12 m\ell^2\dot\theta^2 + mg\ell\cos\theta, \qquad m\ell^2\ddot\theta + mg\ell\sin\theta = 0.$$

In units with $m = g = \ell = 1$ this is $\ddot x = -\sin x$ of (1.12). The rod's tension, which the equation never mentions, follows from the motion, as the theorem says: $F^{\mathrm c} = m\ddot r + \nabla V$.

**Summary.**

- *Handed on:* the configuration space $Q$, a surface of dimension $n$ covered by patches of generalised coordinates with smooth transitions (Corollary 2.4 and Lemma 2.5); its velocities $T_rQ = \ker Df$, which transform by $D\varphi$ (Theorem 2.7, (2.4)).
- *The main result:* for ideal constraints, Newton's law is Lagrange's equations for $\mathcal{L} = T - V$, in any patch, and for $C^3$ data $(q,\dot q)$ determines the motion (Theorem 2.10 and Corollary 2.11).
- *Next:* Lecture 3 asks why the *form* of these equations survives every change of coordinates: they make one number, the action, stationary.

## References

Arnold, *Mathematical Methods of Classical Mechanics*, §§17–18 (constraints and examples, pp. 75–83) and §21 (d'Alembert, pp. 91–97). Tong, *Lectures on Classical Dynamics*, §2.3 (constraints, pp. 17–21).